

JOSEPH C. DILLARD

Mission Systems / Geospatial Platform Architect | Staff-Level Technical Leader

Rockwall, TX | 254-266-5410 | joseph.dillard@outlook.com | linkedin.com/in/joe-dillard-gis-architect | github.com/JosephDillard

PROFESSIONAL SUMMARY

Mission-focused GIS and software architecture leader with 25+ years of geospatial experience, 20+ years designing and operating Enterprise GIS platforms, and 9+ years of Java/GIS web development. Built and led architecture for high-availability situational awareness and common operating picture systems used by Pacific Air Forces and U.S. Forces Korea. Strong fit for mission planning software that requires system design, product judgment, real-time operational data, secure integrations, maps, dashboards, customer discovery, and calm execution during exercises and high-impact events. U.S. Army Veteran; current Security+ CE; experienced in DoD, utility, and enterprise environments.

TARGET ROLE ALIGNMENT

- **Mission planning and C2 software:** Designed and led an ArcGIS-based common operating picture platform supporting commanders, staff workflows, map views, dashboards, operational planning, exercises, simulations, and planning conferences.
- **Staff-level architecture ownership:** Owned architecture direction across infrastructure, GIS services, custom applications, data models, performance, release planning, and operational readiness across multiple major system revisions.
- **Full-stack and platform delivery:** Hands-on experience with Java, JavaScript, Angular, Node.js, Spring Boot, Grails, Python, REST/SOAP APIs, relational/spatial databases, and enterprise service integrations.
- **Operational reliability at scale:** Maintained zero downtime due to technical failure during major use events with up to 2,900 unique users and observed load up to 20,000 ArcGIS Server requests per minute.
- **Ambiguity to product clarity:** Converted unclear product owner and mission stakeholder needs into practical requirements, MVPs, workflows, releases, training, and production-ready systems.
- **Military + technologist bridge:** Combines U.S. Army GEOINT background, 17+ years supporting Air Force/USFK enterprise GIS missions, and senior solution architecture for defense and utility customers.

TECHNICAL SKILLS

Product / Architecture	System design, enterprise architecture, platform modernization, technical direction, use-case discovery, requirements shaping, MVP delivery, architecture tradeoffs, release planning, technical mentoring
Mission / Geospatial	ArcGIS Enterprise, ArcGIS Server, Portal for ArcGIS, ArcGIS Pro, ArcSDE, ArcGIS JS API, Utility Network, QGIS, GeoServer, GeoPandas, maps, dashboards, common operating picture, situational awareness
Full-Stack Development	Java, JavaScript, Angular, Node.js, Python, C#, Spring Boot, Grails, Hibernate/GORM, REST/SOAP APIs, HTML/CSS, Maven/Gradle, Git/GitHub, IntelliJ IDEA, VS Code, Codex
Data / Integrations	PostgreSQL/PostGIS, Oracle Database 11g/12c, Oracle Spatial, MySQL, SQL, PL/SQL, GeoJSON, CSV/Excel data workflows, NiFi, geospatial ETL, API integration, data readiness and quality checks
Infrastructure / Security	Microsoft Azure, AWS working knowledge, Docker, Windows Server, Linux/Unix, VMware vSphere, Hyper-V, Apache Tomcat, enterprise hosting, capacity planning, RMF/security-aware environments, Security+ CE
Delivery / Quality	Agile/Scrum, Jira/Confluence, stakeholder alignment, training, testing, JMeter, Xcode Instruments, customer enablement, production readiness, operational handoff

CERTIFICATIONS

- Microsoft Certified: Azure Solutions Architect Expert
- AZ-305 - Designing Microsoft Azure Infrastructure Solutions
- AZ-104 - Azure Administrator Associate
- CompTIA Security+ CE - current DoD 8570-aligned certification

PROFESSIONAL EXPERIENCE

TRC Companies - Lemur Team

Sept 2023 - Mar 2026

GIS Solutions Architect / Enterprise GIS Consultant

- Consulted with electric utility customers to assess enterprise GIS readiness for Lemur mobile GIS workflows, focusing on reliable ArcGIS Enterprise services, data architecture, field usability, and operational fit.
- Worked across ArcGIS Pro, ArcGIS Server, PostgreSQL/PostGIS, ArcGIS Utility Network, and the Lemur GIS application to support service publishing, integration, and mobile application consumption.
- Reviewed customer Enterprise GIS configurations, service dependencies, data readiness, and performance constraints before deployment into production-oriented mobile workflows.
- Partnered with GIS architects, developers, customer technical teams, and application stakeholders to troubleshoot service, data, workflow, and performance issues affecting deployment readiness.

- Supported training and customer enablement by explaining application workflows, GIS service dependencies, testing procedures, operational handoff needs, and production support considerations.
- Participated in application, service, and workflow testing to validate functionality, performance, data behavior, and field usability for enterprise utility GIS operations.

L3Harris Technologies - Greenville, TX

Aug 2021 - Aug 2023

Scientist, System Engineer - GIS Solutions Architect

- Designed and configured commercial and open-source GIS server systems for internal research and development initiatives supporting defense-oriented geospatial and signal intelligence environments.
- Led use-case and requirements discovery for commercial and internal GIS systems under development, aligning technical architecture with mission needs, data availability, and stakeholder priorities.
- Researched, tested, and evaluated GIS server software for use in signal intelligence system environments, balancing capability, integration fit, operational constraints, and production readiness.
- Coordinated acquisition of GIS datasets from DoD agencies and supported integration into enterprise geospatial systems requiring reliable ingest, transformation, and data management.
- Built automated geospatial data pipelines using NiFi and GeoPandas to convert GeoJSON into Oracle Spatial, reducing manual ingest effort and improving repeatability of data acquisition workflows.

CACI International Inc. / U.S. Forces Korea

Jan 2013 - May 2021

GIS System Architect - GIS Developer

- Architected and led development of an ArcGIS-based situational awareness/common operating picture platform used by Pacific Air Forces and U.S. Forces Korea for mission planning, operational awareness, exercises, simulations, and planning conferences.
- Designed system architecture from inception through four major revisions over eight years, including infrastructure, data services, application patterns, performance strategy, release planning, and support model.
- Supported more than 50 custom applications, map views, and dashboards that combined commander-reported data across echelons into operational products for faster analysis and decision-making.
- Led two major architecture and application releases each year, coordinating requirements, development priorities, testing, deployment readiness, training needs, and stakeholder communication.
- Translated ambiguous mission stakeholder and product owner needs into practical requirements, workflows, MVP-based delivery plans, sprint priorities, and deployable capabilities.
- Provided technical direction and mentoring for system administrators, web developers, GIS administrators, and database-focused team members maintaining ArcGIS services, Oracle data, and custom applications.
- Selected server and network hardware to ensure compatibility, performance, availability, and capacity for projected operational growth across enterprise environments.
- Maintained zero downtime due to technical failure during major use events with up to 2,900 unique users and observed load up to 20,000 ArcGIS Server requests per minute.
- Worked directly with military customers and staff stakeholders during operational cycles to keep map-based products, dashboards, and decision-support workflows aligned with mission requirements.

CACI International Inc. - Osan Air Base, Korea

Feb 2004 - Jan 2013

Senior GIS Analyst - GIS System Administrator

- Administered ArcGIS Server, ArcIMS/IMF, and ArcSDE Oracle database servers across multiple enterprise networks supporting Air Force and U.S. Forces Korea geospatial operations.
- Led architecture changes to system hardware and software configurations and delivered 24/7 technical support for enterprise GIS operations with operational exercise and contingency support requirements.
- Maintained and disseminated geospatial data in support of Air Force Enterprise GIS Architecture and expeditionary GeoReach requirements, keeping mission users supplied with current geospatial information.
- Managed Windows Server and VMware vSphere virtual machines supporting enterprise GIS services, application hosting, database connectivity, and operational support needs.
- Processed commercial and NGA imagery using ERDAS IMAGINE and provided limited Trimble survey support for operational and data maintenance requirements.

U.S. Army

Apr 2000 - Feb 2004

Production Manager and GEOINT Analyst

- Managed mission-critical GIS-driven product development, map production, topographic operations, QA, imagery support, and geospatial information delivery for operational missions.
- Supervised personnel, provided technical guidance, authored standard operating procedures, and trained users on imagery and GIS equipment.
- Produced more than 375 GIS and map-based documents while improving team productivity, collaboration, and access to critical geospatial information.

- Applied imagery analysis, terrain analysis, cartographic production, database management, and quality assurance practices in support of operational users.

SELECTED MISSION / PRODUCT ARCHITECTURE HIGHLIGHTS

- **Mission command / COP architecture:** Led long-running architecture and development for a platform that converted distributed commander-reported information into live map views, dashboards, and operational products.
- **Real-time geospatial user experience:** Built systems around maps, dashboards, structured operational data, and service-backed applications so users could analyze mission information instead of searching for it.
- **High-load enterprise service operations:** Planned capacity and operational resilience for major use events and sustained mission periods where outages would affect staff workflows and decision timelines.
- **Data integration and automation:** Built repeatable geospatial ingestion patterns across GeoJSON, Oracle Spatial, PostGIS, ArcGIS services, and automated pipeline tooling.
- **Customer/stakeholder translation:** Regularly converted vague operational goals into technical scope, workflow designs, and deployable software or GIS service capabilities.

EDUCATION AND PROFESSIONAL DEVELOPMENT

- Microsoft AI Skills Fest - Jun 2026
- Esri Developer Summit - Palm Springs, CA - Mar 2018
- Western Governors University - Web Development coursework - 2012-2013
- National Geospatial-Intelligence College - Geographic Information Sciences - 2000

SELECTED PUBLIC REFERENCE

GeoBase Mission Command System article: dvidshub.net/news/195988/pacaf-utilizes-geobase-program